

Course Code	Course Title	L	R	P	Cr
23ME3000E303	Operations Research	3	0	0	3
Open Elective					

Unit 1 – Linear Programming Problem

Lesson 6

1. Linear Programming
2. LPP Formulation
3. Graphical Method
4. Simplex Method with two variables
5. Simplex Method with three variables
6. Simplex Method for Minimization

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Duality



Minimization problems are transformed into maximization problems (dual problems) using a duality principle.



The dual problem is created by transposing constraint coefficients, swapping objective function coefficients with RHS values, and changing inequality signs.



Duality provides a way to solve minimization problems with the Simplex Method designed for maximization. It can also offer insights into the original problem's structure.

Use simplex method to solve the following linear programming problem,

$$\text{Minimization, } Z = 12X_1 + 16X_2$$

$$\text{Subject to } X_1 + 2X_2 \geq 40$$

$$X_1 + X_2 \geq 30$$

$$X_1, X_2 \geq 0$$

1. Convert Minimization Problem to Maximization

$$Z=12X_1+16X_2$$

$$X_1+2X_2\geq 40$$

$$X_1+X_2\geq 30$$



X_1	X_2	
1	2	40
1	1	30
12	16	0

$$Z=40Y_1+30Y_2$$

$$Y_1+Y_2\leq 12$$

$$2Y_1+Y_2\leq 16$$



Y_1	Y_2	
1	1	12
2	1	16
40	30	0

1. Introduce Slack Variable

$$Z = 40Y_1 + 30Y_2$$

$$Y_1 + Y_2 \leq 12$$

$$2Y_1 + Y_2 \leq 16$$

$$-40Y_1 - 30Y_2 + 0 \cdot S_1 + 0 \cdot S_2 + Z = 0$$

$$Y_1 + Y_2 + S_1 = 12$$

$$2Y_1 + Y_2 + S_2 = 16$$

2. Construct Matrix with Coefficients

$$-40Y_1 - 30Y_2 + 0 \cdot S_1 + 0 \cdot S_2 + Z = 0$$

$$Y_1 + Y_2 + S_1 = 12$$

$$2Y_1 + Y_2 + S_2 = 16$$

	Y_1	Y_2	S_1	S_2	Z	
R_1	1	1	1	0	0	12
R_2	2	1	0	1	0	16
R_3	-40	-30	0	0	1	0

3. Do Iterative Process

	Y_1	Y_2	S_1	S_2	Z	
R_1	1	1	1	0	0	12
R_2	2	1	0	1	0	16
R_3	-40	-30	0	0	1	0

$$R_1 \rightarrow R_1 - R_2/2 \quad R_2 \rightarrow R_2/2 \quad R_3 \rightarrow R_3 + 20R_2$$

	Y_1	Y_2	S_1	S_2	Z	
R_1	0	$1/2$	1	$-1/2$	0	4
R_2	1	$1/2$	0	$1/2$	0	8
R_3	0	-10	0	20	1	320

3. Repeat Iterative Process

	Y_1	Y_2	S_1	S_2	Z	
R_1	0	1/2	1	-1/2	0	4
R_2	1	1/2	0	1/2	0	8
R_3	0	-10	0	20	1	320

$$R_1 \rightarrow 2R_1 \quad R_2 \rightarrow R_2 - R_1 \quad R_3 \rightarrow R_3 + 20R_1$$

	Y_1	Y_2	S_1	S_2	Z	
R_1	0	1	2	-1	0	8
R_2	1	0	-1	1	0	4
R_3	0	0	20	10	1	400

Solution: $X_1 = 20$, $X_2 = 10$, $Z = 400$

Use simplex method to solve the following linear programming problem,

Minimization, $Z=4X_1+6X_2$

Subject to $3X_1+X_2\geq 3$

$$X_1+2X_2\geq 4$$

$$X_1, X_2\geq 0$$

Conclusion

The Simplex Method is a powerful tool for solving linear programming problems, offering solutions for a wide range of applications in various industries.

The Simplex Method is best suited for linear problems. For non-linear optimization, other methods may be more appropriate.

Advanced topics like sensitivity analysis, post-optimality analysis, and integer programming can be explored for a deeper understanding of LP.